

Watching the Watchers:

A Comparative Audit of Cloud-Based Commercial Content Moderation Services

Amin Oueslati (Hertie School)

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Supervisor Prof. Simon Munzert

Practice Partner Weizenbaum Institute
Research Group: Data, Algorithmic Systems and Ethics

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Extended Abstract. In today’s digital landscape, online platforms face the challenge of moderating an ever increasing volume of content, including harmful hate speech. While the extent to which platforms rely on algorithms to moderate speech remains unknown, it seems plausible that most use them at least in an auxiliary function, flagging speech for human moderators, while some might even deploy them fully automatically. Absent a coherent legal or theoretical taxonomy, companies and content moderation algorithms alike are given great levels of discretion when moderating hate speech. Considering such latitude, the lack of transparency and external accountability is particularly alarming.

Our study aims to fill this gap by systematically evaluating four leading commercial cloud-based content moderation services, offered by Amazon, Google, Microsoft and OpenAI, through a third-party audit. On the one hand, we aim to highlight specific issues pertaining to these four services, on the other hand, we hope to establish a test suite that may guide future audits. To these ends, we conduct three experiments. First, we evaluate service performance on three popular hate speech datasets, ToxiGen, Jigsaw and MegaSpeech, which were chosen for their popularity and capacity to capture different forms of hate speech. We compute performance metrics both at the aggregate- and at the group-level, parting from most prior research by extending the group-level analysis to a total of eight minority identities. Second, we further test for group-specific biases by running Perturbation Sensitivity Analyses on the most common identity tokens associated with each of the eight minority groups. In essence, we measure the extent to which content moderation services attach a negative bias to these minority tokens, compared to counterfactual majority tokens. We construct these examples from the *Identity Phrase Templates* in [Dixon et al. \(2018\)](#) and MegaSpeech. While the former is composed of synthetic phrases, designed for Perturbation Sensitivity Analysis, the latter comprises of real-world examples. Third, we deploy a Structural Topic Model to explore substantively which topics characterise phrases that were either over- or undermoderated by all services.

Our analysis reveals substantial issues, some of which are confined to a particular service, while others extend to all services. All services had difficulties detecting implicit hate speech, which relies on more subtle and codified messages. Moreover, our results point to the need to remove group-specific bias. It seems that biases towards some groups, such as *Female*, have been mostly rectified, while biases towards other groups, such as *LGBTQ* and *Black* remain. The results from our Perturbation Sensitivity Analysis suggest that this bias is not primarily driven by greater toxicity attached to common identity tokens, with the exception being group *LGBTQ*. Instead, the bias might be encoded in other terms that routinely occur with these minority groups. Lastly, it appears that developers must consider a broader set of vulnerable groups as potential victims of hate speech. While services perform reasonably well at detecting hate against groups which are routinely covered in hate speech research, hate against groups *Disability*, *Latinx* and *Asian* often goes undiscovered.

1 Background

The digital age has brought a substantial increase in online content. Worryingly, this also includes pernicious, unwanted content, such as hate speech (Bakalis, 2016). Online platforms responded by adopting extensive content moderation regimes (De Gregorio, 2019). Absent a legal taxonomy of what constitutes hate speech, private companies are given great autonomy in their moderation practices, effectively making them the judges of public speech (Seering, 2020; Einwiller and Kim, 2020). Typically, large online platforms rely on so-called community guidelines or terms of service against which speech is assessed. The assessment is done by human moderators, who are assisted by algorithms or, in some instances, fully automated (Gorwa et al., 2020). The largest technology firms, such as Google, Microsoft, and Amazon, additionally offer content moderation as a service via cloud-based API access. While most organisations do not report the extent to which algorithms shape their content moderation, the sheer amount of online speech makes reliance on algorithmic moderation inevitable (Schluger et al., 2022).

The Perils of Algorithmic Content Moderation

The performance and non-discrimination of content moderation algorithms matters, since digital activity in general, and online speech in particular, play a decisive role in modern society. If content moderation algorithms malfunction, some users are wrongfully censored, while others are insufficiently protected (Dixon et al., 2018). To date, there exists no systematic evaluation of commercial cloud-based content moderation services, a gap this paper aims to fill. The absence of public scrutiny is alarming, as many open-source content moderation algorithms have been tied to problematic performance and biases. First, hate speech detection algorithms regularly suffer from poor generalisation, as the training data heavily oversamples certain users. For instance, 70% of all sexist tweets, and 90% of all racist tweets in a hate speech dataset by Waseem and Hovy (2016) belonged to a single author (Wiegand et al., 2019). Second, research revealed overmoderation of neutral statements mentioning vulnerable groups (Sheng et al., 2019). This is fueled by the historic discrimination of vulnerable groups, which often makes them the target of hate speech (Wiegand et al., 2019). Algorithms

internalise such spurious correlations as they occur in the training data. Third, reappropriated language has been linked to algorithmic discrimination of minorities. Reappropriation is a sociocultural process where oppressed groups transform previously derogatory language into expressions of their own empowerment, with group members using on the outset disparaging terms when referring to themselves or other group members (Galinsky et al., 2003). For instance, Sap et al. (2020) find that surface markers of *African-American-English* are stronger correlated with toxicity than what the authors call *White-Aligned-English*. It is worth noting that prior research concerning biases in content moderation algorithms heavily focused on women and people of colour, while the discrimination of other groups, like *LGBTQ* people, people with *Disabilities* or *Latinx* was addressed to a much lesser extent (Garg et al., 2023). Similarly, most research probed the performance of hate speech moderation algorithms on English. Tests on non-English languages pose the exception and include for instance Hindi in Ghosh et al. (2021) or Arabic in (Khezzar et al., 2023). These studies often find that content moderation algorithms perform poorer on non-English languages.

The Role of Third-Party Auditing

Audits are critical for ensuring the functioning and accountability of algorithmic systems (Raji and Buolamwini, 2019; Bandy, 2021). Independent third-party audits, as pursued by this paper, are of particular importance, as they are associated with greater scrutiny than internal (first-party) or contracted (second-party) audits (Birhane et al., 2024; Raji et al., 2022). Given their greater levels of independence, third-party auditors can probe harshly and publish critical results, mostly without interference from the companies. On the flip side, the unsolicited nature of third-party audits also constrains access. Ideally, auditors have full, so called *white-box* access which allows for the most comprehensive assessment, including tweaking model weights, re-training and fine-tuning (Casper et al., 2024). Absent such access, third-party audits must develop strategies to scrutinise the algorithm albeit limited *black-box* access. Some recent political developments signal commitment to enabling systematic third-party audits. More specifically, the EU Digital Service Act (DSA) Art. 40 establishes a right for vetted researchers to access the data of Very Large Online Platforms and examine questions related to systemic risks (European Parliament, 2022). While the goal is laudable,

its efficacy remains to be seen. For instance, it seems likely that companies constrain access by invoking the protection of their trade secrets ([Leerssen, 2021](#)). Additionally, in some cases, such as Google Jigsaw, content moderation services are offered by a spin-off from the parent company. While Google is a designated Very Large Online Platforms and required to offer researcher access, Jigsaw is not ([European Commission, 2024](#)).

Contribution

This paper’s contribution is twofold. First, it offers the first comprehensive assessment of four major commercial cloud-based content moderation algorithms. Not only are these algorithms likely in use at the companies themselves, which, taken together, amount to billions of users, but they are also likely deployed by a vast number of smaller organisations through the SaaS model. For instance, Jigsaw’s Perspective AI has been deployed by the New York Times, Vox Media and OpenWeb, among others, and processed 500M requests daily in 2021 ([Newswire, 2021](#)).

Second, this paper’s auditing strategy may inform future black-box audits of cloud-based content moderation algorithms. While initiatives like the DSA might provide enhanced access to major platforms, unsolicited black-box audits likely maintain their importance for reasons discussed. We design a suite of tests to scrutinise cloud-based moderation services both at the aggregate- and at the group-level, across several datasets. At the group-level, we part from most prior research by examining biases against a broader set of vulnerable groups.

2 Data and Method

Our research team gained API-access to Google Moderate Text, Amazon Comprehend, Microsoft Azure Content Moderation, and OpenAI Content Moderation. Rather than returning a single score, these services typically report hate speech along several sub-categories, such as *offensive*, *insult* or *profanity*. For the purpose of this paper, we treat the maximum value among sub-categories as a statement’s overall *toxicity score*, as all sub-categories pertain to forms of speech most community guidelines and terms of service would ban. While the relevant literature uses diverging terminology, the term *toxicity score* is commonly adopted when describing aggregate levels of hate speech (Fortuna et al., 2020). When a phrase is linked to a toxicity score greater than 0.5, we classify it as hate speech. Absent definite guidance from the service providers, we consider these realistic specifications for deployment. Further details are presented in table A1 in the appendix. We experimented with different deployment specifications, which did not substantially alter the paper’s main findings.

2.1 Experiment 1: Performance on Popular Hate Speech Datasets

We use three datasets, MegaSpeech, Jigsaw, and ToxiGen (Pendzel et al., 2023; Jigsaw, 2019; Hartvigsen et al., 2022). These datasets were selected for their popularity and capacity to capture various common forms of hate speech. ToxiGen primarily contains implicit hate speech, which is constructed around indirect and codified messages, making it more difficult to detect (ElSherief et al., 2021). By contrast, explicit hate speech is characterised by the use of overtly offensive language. MegaSpeech and Jigsaw offer a mix of explicit and implicit speech. Moreover, MegaSpeech and ToxiGen leverage generative AI to scale and diversify speech corpora. This approach creates greater lexical variation in the data, probing the algorithms’ ability to generalise (Pendzel et al., 2023; Hartvigsen et al., 2022). ToxiGen ensures the robustness of its machine-generated identity and hate labels by using demonstration-based prompting and subsequent review from human annotators. MegaSpeech uses a prominent hate speech classification algorithm to assign hate labels and discards any classifications below a 0.7 certainty threshold. It does not come with identity labels. Jigsaw, on the other hand, solely contains human-written examples, labelled by human annotators. Structurally,

MegaSpeech and ToxiGen provide shorter text sequences, with on average 19.0 and 18.2 words respectively, while Jigsaw is made up by longer phrases, 47.6 words on average. Table A2 in the appendix offers a descriptive overview of the three datasets.

All cloud-based moderation algorithms are evaluated across all datasets on a set of threshold-variant and threshold-invariant performance metrics (Elsafoury et al., 2023; Borkan et al., 2019). More specifically, we compute a scale-variant F1 score, True Positive Rate and False Positive Rate, as well as a scale-invariant ROC AUC Score. A key aspect is the calculation of these metrics not only at the aggregate level but also specifically for vulnerable groups, with the goal of uncovering any group-level variation that might indicate discrimination. Assuming similar performance metrics across groups reflects an *Equality of Odds* notion of fairness, as theoretically proposed in Hardt et al. (2016) and empirically implemented in Dixon et al. (2018). At the group-level, we compute the pinned ROC AUC, a novel metric proposed by Dixon et al. (2018), designed to provide a more robust measure for scale-invariant performance comparison across sub-groups. Essentially, it *pins* the data of each sub-group to the same underlying distribution by computing the ROC AUC on a dataset which is made up of 50% data pertaining to the sub-group and 50% data randomly drawn from the full dataset. While this approach also comes with its pitfalls, as the authors themselves note in a subsequent paper, it is the best scale-invariant metric to date when presented with group-level variation in biases (Borkan et al., 2019).

For Jigsaw, target groups and hate labels were coded by human annotators. For ToxiGen we constrain our analysis to a subset of 10,000 observations, which was annotated by human reviewers. For both Jigsaw and ToxiGen, when running aggregate-level tests, we only include phrases which were assigned the same hate label by all annotators. For group-level analysis, we additionally require consensus on the target group’s identity. Hate labels in MegaSpeech were assigned by a prominent hate speech classifier, and a random sample validated by human annotators. Unfortunately, these annotations are not made accessible. To assign identity labels to MegaSpeech, we trained a BiLSTM identity classifier, exploiting a relevant dataset by Yoder et al. (2022). The classifier achieves an accuracy of 78% on a hold-out test dataset. In total, we explore biases against eight minority groups, including (*Asian, Black, Disability, Female, Jewish, Latinx, LGBTQ, Muslim*), which can be mapped across all

datasets with limited assumptions. All datasets are balanced on toxic and non-toxic speech, both at the aggregate and the group-level, to avoid distortion of performance metrics.

Limitations and Future Work

A key limitation relates to the inherent subjectivity of hate speech. We assess cloud-based moderation services against the ground truth in the data, which was established or validated by human annotators. Hence, when we say a particular service moderated a phrase incorrectly, strictly speaking, we mean that it deviates from the consensus among human annotators. To the extent that citizen preferences may inform hate speech moderation, as proposed for instance by [Munzert et al. \(n.d.\)](#), examining such deviations seems meaningful. Naturally, whenever a service systematically deviates between datasets, we treat such patterns as stronger evidence for algorithmic deficiencies. Moreover, we inspect the comparative moderation of services on the same dataset. Again, when one service substantially and persistently deviates from the other services, this may indicate service-specific problems.

In the face of these limitations, we see an urgent need for future research to establish an overarching taxonomy of hate speech ([Fortuna et al., 2020](#)). Such a taxonomy must assert consistent definitions for the various concepts that are often subsumed under hate speech, such as offensive, derogatory or insulting speech. Ideally, the taxonomy would also offer guidelines whether and under which conditions such concepts should be deemed hate speech and requires moderation. Given the normative nature of such a taxonomy, we believe its development necessitates an inclusive process involving a broad range of stakeholders, such as the affected groups, legal experts and representatives of broader society. Benefits of the taxonomy would be multifaceted. It would enable consistent annotation schemes, guide developers when designing moderation algorithms and establish a baseline against which auditors may assess them.

Moreover, while group-level fairness, or the lack thereof, is a primary motivation for this research, its assertion is outside our work’s scope. Instead, our focus is mainly descriptive in nature, aimed at uncovering group-level variations in moderation services’ performance. Translating these descriptive measures into normative judgements is a challenging endeavour as it is highly debated what would constitute truly fair outcomes. While some argue

for equalising group-level outcomes, others demand affirmative action which might grant historically oppressed groups a stronger right to assert themselves in society, which might coincide with less stringent moderation of their speech - see for instance [Binns \(2020\)](#) for an overview.

Lastly, all datasets on which we evaluate commercial cloud-based content moderation services are in English. Worryingly, high-quality non-English hate speech datasets are rare, a shortfall future foundational work must address. Additionally, we are constrained by the fact that no one in our research team fluently speaks a major non-English language such as Mandarin, Hindi, Arabic, or Spanish.

2.2 Experiment 2: Perturbation Sensitivity Analysis

Perturbation Sensitivity Analysis offers an additional, arguably more robust evaluation of group-level biases. In Experiment 1, we derive conclusions about group-level biases from variation in False Negative and False Positive Rates. This makes two assumptions. First, the validity of the ground truth in the data. Paradoxically, an *unbiased* moderation service would appear *biased* if the bias is actually encapsulated in the data itself. Second, we assume that data on all groups comes from the same underlying distributions. If one group were to contain many more edge cases than other groups, this exacerbates False Positive and False Negative Rates, compromising our ability to infer biases.

Perturbation Sensitivity Analysis avoids such pitfalls by solely exchanging group-specific identity tokens while holding the remainder of the sentence constant ([Prabhakaran et al., 2019](#)). Table 1 displays the fundamental logic. We follow prior research in defining an anchor group against which other groups are compared ([Prabhakaran et al., 2019](#)). Using the dominant majority group as baseline, Counterfactual Token Fairness scores are computed as the difference in toxicity between the baseline and the corresponding minority group.

Perturbation Sensitivity Analysis makes two assumptions. First, that the majority and minority counterfactuals convey the same (neutral) meaning. Conceptually, this requirement seems met if we compare *black* to *white*, but violated if we were to compare *ni**** to *white*. While constructing toxic counterfactuals is theoretically possible, it is methodologically demanding and exceeds the scope of this project. Instead, we construct 34 *neutral*

counterfactual pairs which are reported in table A3 in the appendix. Importantly, each minority group is represented by multiple tokens, reflecting its different semantic representations. For instance, the minority group *female* also manifests as *woman* and *women*. Perturbation Sensitivity Analysis’s second assumption requires there to be no unique interaction between a particular minority token and the underlying sentence. This *symmetric counterfactual* assumption is regularly violated in real-world cases, for instance, in the presence of stereotypes (Garg et al., 2019). Take the sentence *<IDENTITY> folks have weird eyes*. Attaching greater toxicity to *Asian folks have weird eyes* than to *White folks have weird eyes* appears intuitively correct, since the former picks up on a particular discriminatory stereotype. To satisfy this second assumption, we choose data that largely contains short sentences and explicit hate speech, two features which limit the occurrence of stereotypes and other semantic peculiarities (Garg et al., 2019). Additionally, as proposed by Garg et al. (2019), Counterfactual Token Fairness scores are computed separately for toxic and non-toxic statements. Counterfactual symmetry holds more robustly for the latter.

Experiments are conducted using two distinct datasets. First, the synthetic *Identity Phrase Templates* from Dixon et al. (2018) are used. The set contains 77,000 synthetic examples of which 50% are toxic. These avoid stereotypes and complex sentence structures by design, which ensures that the symmetric counterfactual assumption is met. Mapping the dataset, which contains a broader set of identities, to the 34 minority token relevant to this study, results in 25,738 sentence pairs. Their distribution across minority groups is reported in appendix table A4. Additionally, table A5 in the appendix illustrates their semantic style through select examples. Second, by applying the same logic, 9,190 sentence pairs are derived from the MegaSpeech dataset. Their distribution is presented in table A4 in the appendix, while table A6 displays select examples. MegaSpeech is chosen as it contains short sentences and primarily uses explicit hate speech. While the synthetic dataset consists of extremely stripped down phrases, which allow us to truly isolate variation in toxicity, the non-synthetic dataset tests the extent to which patterns persist when extended to more realistic instances of speech.

Minority Reference	Toxicity	Majority Reference	Toxicity	PSA
Being Asian is fun	0.21	Being White is fun	0.33	0.12
Being Black is fun	0.31	Being White is fun	0.33	0.02
Being Female is fun	0.16	Being Male is fun	0.18	0.02
Being LGBTQ is fun	0.31	Being Straight is fun	0.25	-0.06
Being Jewish is fun	0.40	Being Christian is fun	0.36	-0.04
Being Muslim is fun	0.42	Being Christian is fun	0.36	-0.06
Being Latinx is fun	0.24	Being White is fun	0.33	0.09

TABLE 1: In Perturbation Sensitivity Analysis, valid counterfactuals are constructed by solely exchanging minority and corresponding majority tokens in a sentence. Counterfactual Token Fairness scores are computed by taking the difference in toxicity score, revealing any groups-specific biases. In this illustrative example, we report the mean toxicity score across all moderation services. Negative Counterfactual Token Fairness scores indicate potentially negative biases associated with a minority’s common identity tokens.

Limitations and Future Work

Our analysis is naturally constraint by the selection of minority tokens. Clearly, minority identities can also manifest in more ways than we cover. For instance, we might link group *Female* to others tokens such as *Girl*, *Wife* or female versions of occupations like *Waitress*. To this end, also the *Identity Phrase Templates* from [Dixon et al. \(2018\)](#) do not offer more manifestations for each identity. We see a need for future work to create larger, more versatile synthetic datasets for Perturbation Sensitivity Analysis.

Moreover, our analysis exclusively focuses on examples where minorities are the target of speech. However, some of the biases uncovered by prior research, such as the overmoderation of reappropriated language, pertains to speech where the minority is the sender. For instance, when minority representatives use language, that is generally perceived derogatory, to refer to themselves or other group members in an empowering way. Such an extension would be meaningful both to synthetic and non-synthetic data.

Lastly, Perturbation Sensitivity Analysis can inherently only test for biases in relation to identity tokens. Importantly however, algorithmic biases can also arise from other words that get uniquely associated with a particular minority. Within the field of Machine Learning,

this relates to the notion *Proxy Discrimination*, where an algorithm uses certain proxies to reconstruct a hidden, protected attribute (Tschantz, 2022). In this vein, research could determine words that co-occur frequently and exclusively with minority tokens, to then construct counterfactual examples which remove or replace these identity proxies.

2.3 Experiment 3: Comparative Structural Topic Model

While the purpose of Experiments 1 and 2 is to uncover deficiencies in commercial cloud-based content moderation services, Experiment 3 aims to explore substantive patterns in moderation outcomes. *Which themes characterise sentences that get overmoderated and undermoderated?* Without asserting a truly causal relationship, such analysis represents a first important step towards correcting any shortcomings that were revealed.

To this end, a Structural Topic Model is developed to explore distinct topics associated with each moderation outcome. Structural Topic Models’ major advantage is the capacity to explicitly model the prevalence of topics as a function of documents’ covariates (Roberts et al., 2019). We estimate topics through the following functional form.

$$prevalence \sim moderation_outcome \times minority_group + dataset$$

Moderation Outcome assesses a service’s prediction against the ground truth. It is a discrete variable with three levels: Undermoderated, correctly moderated, and overmoderated.

Minority Group contains labels for the eight vulnerable groups examined.

Dataset is a discrete identifier recording the origin dataset.

With this functional form, we allow the prevalence of topics to vary by moderation outcome, with the aim of capturing systematic over- or undermoderation. Overmoderation describes a case where a service classifies a phrase as toxic, while the ground truth maintains it is non-toxic. The opposite is true in instances of undermoderation. Additionally, the functional form controls for group- and dataset specific variation in topics. We apply the Structural Topic Model to sentences which were consistently moderated across services. That is to say,

phrases which all services over- under- or correctly moderated. Thereby, we can hone in on those cases which all services classified incorrectly, aiming to uncover potentially systematic issues. Further descriptive statistics on the dataset can be found in appendix table [A7](#).

In total, the corpus consists of 18,955 documents, where each document represents one unique phrase, from either MegaSpeech, ToxiGen or Jigsaw, that was consistently moderated by all four services. Standard pre-processing is applied to the documents, which inter alia involves the removal of special characters, URLs, emojis and empty space. All words are transformed to lower case. Additionally, all identity tokens, as presented in [A3](#), are removed from the corpus, to avoid producing topics which exclusively consist of these identity markers, a behaviour we encountered when running the model without their removal. Lastly, when creating the document-feature matrix, tokens that solely occur once are removed to reduce sparsity.

In topic modelling, the choice of K , the number of topics assigned to the corpus, is determined a priori and somewhat arbitrarily, without a clear optimum. Instead, the selection of K can be guided by statistical methods, but ultimately depends on the underlying research question. This paper employs a two-step approach to find the appropriate K . First, we short-list four promising values K : $K20$, $K30$, $K40$ and $K50$. If the aim was to categorise online hate speech, such high values of K would be unproductive. However, as we are primarily interested in isolating topics which are uniquely associated with over- or undermoderation, the set of relevant topics is naturally reduced, justifying this choice ([Weston et al., 2023](#)). Next, their fit is statistically compared along exclusivity, variational lower bound, residuals and semantic coherence, displayed in figure [A1](#) in the appendix. Second, $K30$ and $K50$ are selected for further substantive analysis, which assesses the extent to which they produce meaningful topics. To this end, we examine key words and key phrases, taking into account probability, frex and lift ([Roberts et al., 2019](#)). Based on this assessment, we decide to conduct subsequent analysis with a $K30$ topic model. Figure [A2](#) in the appendix depicts these 30 topics by prevalence.

This design allows us to statistically test which of the 30 topics are more prevalent with the over- or undermoderation of speech, compared to its correct moderation. This is a core functionality of the Structural Topic implementation in R by [Roberts et al. \(2019\)](#) which

not only statistically estimates the relationship between covariates and topics, but computes the uncertainty associated with these estimates. Employing traditional statistical methods, we identify topics which are more prevalent with over- or undermoderation than correct moderation at an alpha level of 0.05.

Limitations and Future Work

Topic modelling is as much art as it is science. Various decisions, such as the choice of K or their substantive interpretation, introduce a degree of arbitrariness that is inherent in topic models (McFarland et al., 2013). In defense, we do not treat the patterns revealed via topic modelling as definite answers, but rather as a starting point for further inquiry concerning model flaws.

Moreover, we expect unique patterns relating topics and false moderation at the dataset- and the group-level. For instance, it seems likely that undermoderated phrases concerning *Black* people are characterised by different topics than those aimed at *Jewish* people. In theory, a Structural Topic Model would allow to statistically test for such effects, but requires a different methodological design and substantial data at the group-level. However, to develop a better understanding of the causes underlying moderation biases, such analysis will be essential, pointing to an important area of future work.

2.4 A Note on Performance and Efficiency

Auditing cloud-based content moderation algorithms can be demanding, both computationally and financially. Probing these services through different, often explorative tests, can quickly involve millions of API calls. To reduce excessive waiting times and risks of timeout, we parallelise all calls. For the OpenAI Content Moderation API we additionally use batches. Overall, this reduces runtime substantially and enhances the audit’s efficacy and flexibility.

Another obstacles relates to the financial cost of issuing API calls. While a single call costs maximum 0.001\$, these expenditures quickly add up, posing a particular challenge to resource-constraint civil society organisations (Yan and Zhang, 2022; Singh et al., 2023)).

While this paper’s solution was to run analyses on smaller subsets, future work by our wider research group aims to develop active sampling strategies to ensure query-efficient auditing. Active sampling is designed to efficiently select those cases which challenge a service. Put simply, this approach classifies data in smaller batches, identifies incorrectly classified phrases and iteratively retrieves similar instances from the full dataset. A fraction of the next batch is then comprised of these retrieved phrases. Further sampling rules can ensure that the sampling algorithm explores the whole data space. Such reinforced sampling strategies aim to achieve maximal scrutiny with limited resources.

3 Results

3.1 Experiment 1: Performance on Popular Hate Speech Datasets

3.1.1 Aggregate-Level Performance

Dataset	Moderation Service	ROC AUC	F1	FPR	FNR
ToxiGen	Amazon	70.4%	68.9%	7.2%	52.0%
	Google	62.7%	62.7%	39.1%	35.5%
	OpenAI	70.3%	68.1%	33.2%	56.0%
	Microsoft	59.8%	57.4%	16.4%	64.0%
Jigsaw	Amazon	92.2%	92.2%	7.5%	8.1%
	Google	69.9%	67.2%	58.4%	1.8%
	OpenAI	78.6%	78.6%	17.1%	25.6%
	Microsoft	75.8%	75.7%	20.4%	28.1%
MegaSpeech	Amazon	72.8 %	72.0 %	10.4 %	43.9 %
	Google	73.3 %	72.3 %	41.3 %	12.0 %
	OpenAI	77.1 %	76.7 %	8.4 %	37.3 %
	Microsoft	70.6 %	70.1 %	16.9 %	41.9 %

TABLE 2: Performance metrics by moderation service and dataset. The names of moderation services were abbreviated for better readability. ROC AUC is threshold-invariant, while F1, False Positive Rate and False Negative Rate are threshold-variant. Good performance maximises ROC AUC and F1, and minimises False Positive and False Negative Rates. Per dataset, blue shading signals the best performance, while red shading indicates the worst performance. ToxiGen includes 7,800 observations, while Jigsaw and MegaSpeech each contain 50,000. All datasets are balanced on toxic and non-toxic phrases.

Table 2 shows aggregated performance results for Jigsaw, ToxiGen and MegaSpeech, indicating significant disparities between services. Overall, Amazon Comprehend performs best, achieving the best results on ToxiGen and Jigsaw. However, the service’s near optimal performance on Jigsaw (92.2 % ROC AUC) calls for caution, suggesting that the dataset, in full or in parts, was included in Amazon Comprehend’s training data. By contrast, Microsoft Azure Content Moderation performs poorest, achieving the worst results on ToxiGen and MegaSpeech. Notably, Google Text Moderation has the highest False Positive Rate across all datasets, which implies that it tends to falsely classify neutral speech as hate speech, what we conceive as overmoderation. As a natural consequence, Google Text Moderation

is also associated with the lowest False Negative Rate across all datasets, suggesting that it misses very few instances of hate speech.

Lastly, we present some tentative observations regarding cross-dataset performance variation, which however must be treated with great care, given the previously discussed concerns related to inconsistent annotation schemes and the absence of an objective ground truth. When treating Amazon Comprehend’s performance as an outlier, the range of performances seems similar for Jigsaw and MegaSpeech, while somewhat worse for ToxiGen. This disparity seems to manifest as a high False Negative Rate across all moderation services on ToxiGen. One explanation might relate to the prevalence of primarily implicit hate speech in ToxiGen. Rather than using overtly offensive language and slurs, hate speech in ToxiGen is by design masked in subtle and codified messages. To offer an example: *there is a direct correlation between the amount of melanin in a persons skin and how much they are worth as a person* (ID TG3189). Already in the original paper, the authors observe that many open-source hate speech moderation algorithms underperform on ToxiGen (Hartvigsen et al., 2022). Interestingly, the on average much longer text sequences in Jigsaw appear to have no substantial effect on model performance.

3.1.2 Group-Level Performance

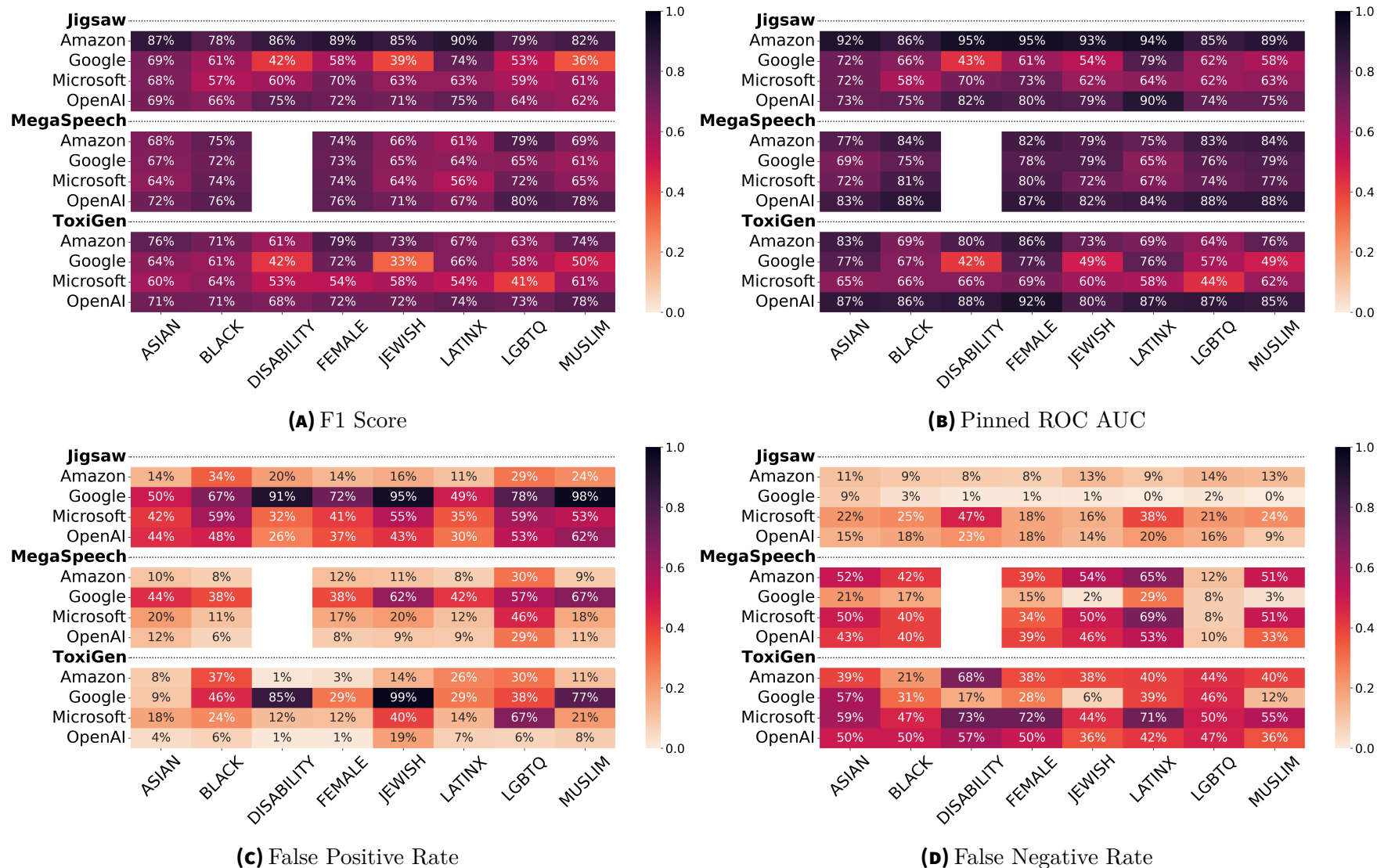


FIGURE 1: Performance metrics by moderation service, dataset and minority group. Moderation services' names were abbreviated for better readability. Pinned ROC AUC is threshold-invariant, while F1, False Positive Rate and False Negative Rate are threshold-variant. Good performance maximises Pinned ROC AUC and F1, and minimises False Positive and False Negative Rates. ToxiGen includes 4,268 observations, Jigsaw consists of 19,228 observations and MegaSpeech is comprised of 33,886. All datasets are balanced on toxic and non-toxic phrases. Group *Disability* was not available for the identity classification of MegaSpeech.

Figure 1 displays performance metrics at the group-level, pointing to notable divergences. Across datasets and moderation services, group *Female* is generally moderated the most accurately, reflected in high F1 and Pinned ROC AUC scores, followed by group *Black*. By contrast, phrases targeting groups *LGBTQ*, *Disability* and *Jewish* overall appear to get moderated the most inaccurately. Interestingly, on MegaSpeech we observe much better performance for group *LGBTQ* by all moderation services. At the service-level, Amazon Comprehend performs consistently well across all minority groups, compared to the other services, while OpenAI Content Moderation ranks second. Microsoft Azure Content Moderation, similar to the results from table 2, achieves the worst results for most minority groups. Strikingly, Google Text Moderation extremely underperforms on groups *Disability* and *Jewish*, compared to other groups, particularly on Jigsaw and ToxiGen.

Inspecting False Positive and False Negative Rates allows for a more substantive interpretation of group-level differences. A high False Positive Rate implies that speech addressing a particular group is overmoderated. Reversely, a high False Negative Rate suggests that hate against a groups goes undetected. Across all services and datasets, groups *Black* and *LGBTQ* by and large are associated with the highest False Positive Rate, resulting in the overmoderation of speech concerning these groups. At the service-level, False Positive Rates for Google Text Moderation are high in general, but extremely high for groups *Jewish Disability*. For instance, on ToxiGen Google Text Moderation produces a False Positive Rate of 99%. Put more tangibly, while the data contains 200 toxic and 200 non-toxic examples for group *Jewish*, Google Text Moderation predicts toxic speech in 385 instances. By contrast, OpenAI Content Moderation, particularly on MegaSpeech and ToxiGen, shows little inclination for overmoderation.

When inspecting False Negative Rates, we immediately notice that moderation services across all groups tend to miss ToxiGen’s implicit hate speech, echoing our prior observations at the aggregate level. For all moderation services apart from Google Text Moderation, speech aimed at group *Disability* seems to be undermoderated the most. Importantly, our results also suggest that comparatively large parts of hate speech targeting groups *Asian* and *Latinx* might go undetected, particularly for Microsoft Azure Content Moderation.

Discussion

Our findings point to concerning flaws of commercial hate speech moderation services, both at the aggregate and the group-level. Importantly, these flaws do not occur uniformly, with some services, such as Amazon Comprehend and OpenAI Content Moderation, performing better than others.

We want to focus our discussion on four patterns we deem particularly concerning and believe require focused attention from developers.

First, all services struggle to detect implicit hate speech, reflected in their high False Positive Rates on ToxiGen. To this end, commercial moderation services do not fare much better than their open-source counterparts (Hartvigsen et al., 2022). One likely cause is the limited availability of implicit hate speech datasets for training purposes. **Second**, all services tend to overmoderate speech concerning groups *Black* and *LGBTQ*. This is somewhat surprising as extensive prior research uncovered biases in open-source content moderation algorithms in relation to these groups (Garg et al., 2019). Commonly, such overmoderation occurs as toxic speech in relation to these groups is overrepresented in the training data, and subsequently learnt by the model. **Third**, most services fail to reliably detect hate speech aimed at groups *Disability*, *Asian* and *Latinx*. Compared to other vulnerable groups, their oppression is less frequently discussed in the hate speech literature (Garg et al., 2023). Consequently, their inclusion in labelling schemes and training datasets might not have been priority to developers. **Fourth**, the tendency of Google Text Moderation to overmoderate is puzzling but also alarming. While we cannot entirely rule out an error on our end, this observation is robust to different configurations of the API. Potentially, this behaviour is intended by its developers. In cases where the algorithm is used in tandem with a human annotator, and punitive action requires human approval, a high False Positive Rate seems preferable compared to a high False Negative Rate. After all, human moderators can always decide not to act on the service’s recommendation. However, in the presence of well-documented automation biases, which entice humans to overly trust algorithmic recommendations, see for instance (Lachemaier et al., 2024), as well as the

possibility of fully automated deployment, recalibrating Google Text Moderation appears necessary.

3.2 Experiment 2: Perturbation Sensitivity Analysis

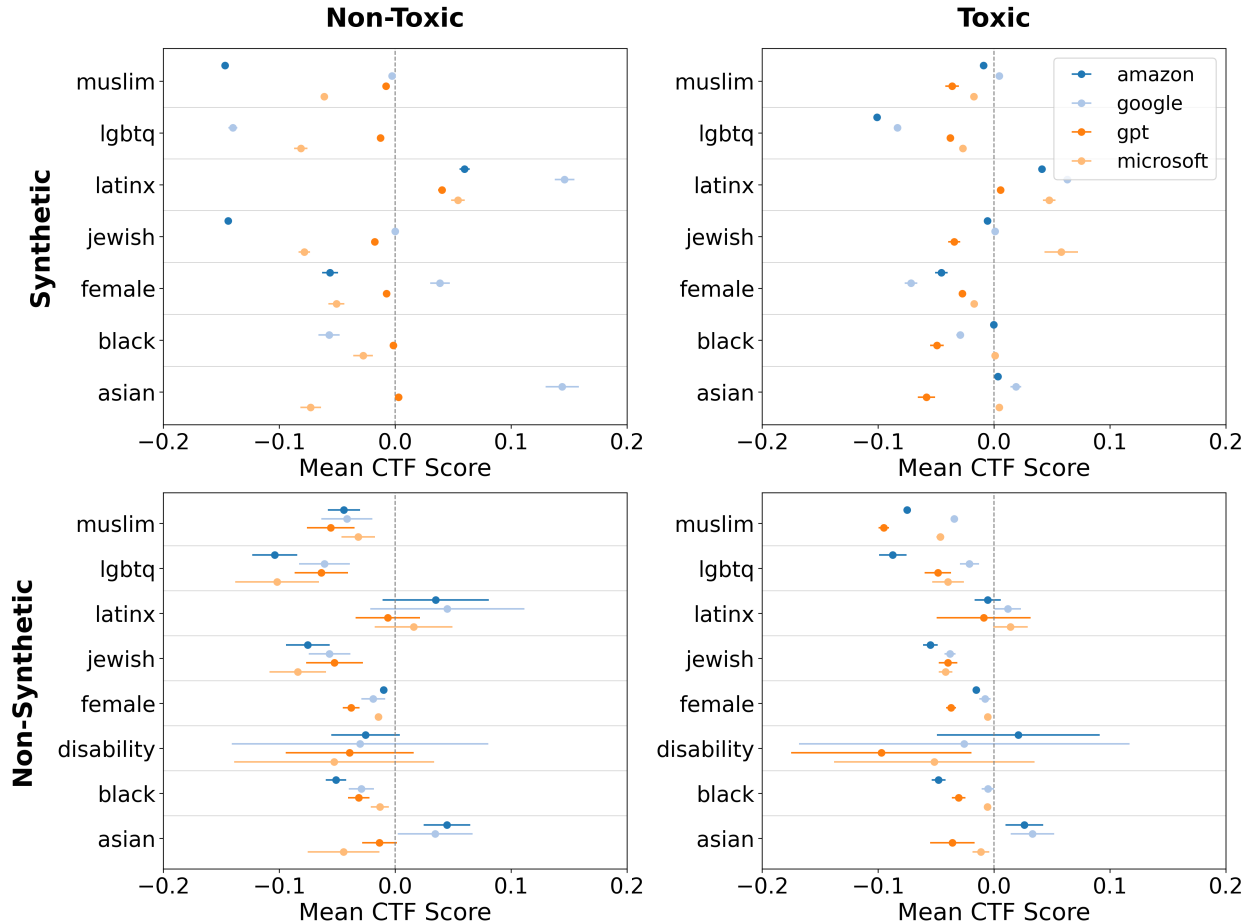


FIGURE 2: Perturbation Sensitivity Analysis on synthetic data from the Identity Phrase Templates in [Dixon et al. \(2018\)](#) and non-synthetic data from MegaSpeech. Counterfactual Token Fairness (CTF) scores are computed as the difference in toxicity between the phrase containing the baseline majority token and its minority perturbation. Counterfactual Token Fairness scores per minority group and service are averaged and reported separately for non-toxic and toxic examples. Besides a point estimate, the figure also includes a 95% confidence interval assuming a student-t distribution.

Figure 2 displays the Perturbation Sensitivity Analysis results on the synthetic and non-synthetic datasets, created from the [Dixon et al. \(2018\)](#) Sentence Templates and MegaSpeech respectively. A negative Counterfactual Token Fairness score indicates that, on average,

minority tokens are associated with higher toxicity scores than their counterfactual majority tokens. Upon general inspection, two observations arise. First, differences in toxicity scores by and large are more pronounced on non-toxic than toxic data. Intuitively this makes sense, as scores are generated non-linearly with a definite upper bound. Thus, when other elements in a sentence induce high toxicity score, the marginal effect from identity tokens is comparably lower. Second, we notice greater variation in the mean Counterfactual Token Fairness scores in non-synthetic than in synthetic data. This was to be expected, as the sentences from MegaSpeech contain more contextual information which interacts with the tokens. Depending on the context, the difference in toxicity between majority and minority token thus varies to a greater extent. By contrast, within non-synthetic examples, the greater variation of Counterfactual Token Fairness mean scores associated with non-toxic data is likely an artefact of the former’s smaller sample size. This observation in itself is revealing: Applying a vector of neutral minority tokens to the MegaSpeech dataset returns a subset of phrases that to 85% consists of toxic phrases.

Overall, the results suggest that most minorities get associated with higher levels of toxicity than dominant majorities, although these effects appear relatively small, and vary across groups and services. Group *LGBTQ* seems associated with the strongest negative bias, occurring for all samples and services. By contrast, we observe limited negative bias against groups *Latinx* and *Asian*. In fact, in most samples and services, minority tokens related to group *Latinx* incur on average lower toxicity scores than their *White* counterfactual. Similarly, groups *Female* and *Black* are linked to comparably small negative biases. Unfortunately, our results allow for no robust inference for group *Disability*. While group *Disability* is entirely absent from the synthetic data, as no examples are included in the (Dixon et al., 2018) Sentence Templates, the non-synthetic data only contains very few instances, resulting in large variation for mean Counterfactual Token Fairness scores.

Other patterns emerge which are more confined to particular services. For instance, Amazon Comprehend is associated with the overall strongest negative bias, linked to groups *Muslim* and *Jewish*. Notably, when applied to toxic data, this effect vanishes. Interestingly, for Google we are presented with findings that are somewhat difficult to reconcile with the previous section’s result. More specifically, we would have expected a large negative

bias placed by Google Text Moderation onto group *Jewish*, which does not seem to be the case. Potentially, the extreme False Positive Rates were not driven by *Jewish* identity terms themselves, but rather by other words co-occurring, such as *antisemitic* or *Israel*. Further research is required to explain the disparity observed. Lastly, it is worth noting that OpenAI Content Moderation seems the least biased, across groups and datasets.

Discussion

Among all challenges to group-level fairness, spurious correlations attached to common identity tokens are in our opinion among the simplest to rectify. While prior work, such as [Madhavan et al. \(2023\)](#), found that open-source hate speech detection algorithms are negatively biased against common identity tokens, we largely fail to reproduce these with major commercial services. Reversely, this implies that many of the flaws we uncovered during Experiment 1 in fact relate to more complex dynamics, which probably makes them also more difficult to rectify.

Nevertheless, we want to highlight two bias patterns, we believe developers should address.

First, some negative biases on common identity tokens remain, most consistently linked to group *LGBTQ*. Upon initial inquiry, it seems much of this penalty is linked to group *Gay* within the umbrella term *LGBTQ*. On average, *Gay* is linked to a mean Counterfactual Token Fairness Score of -0.24 on the synthetic data, whereas *LGBT* is only associated with a score of -0.05. This result is surprising, given that discrimination towards group *Gay* is not novel but well-established. Likely, sentences containing the term *Gay* in training datasets are overwhelmingly labelled as hate speech. Nevertheless, we would have expected developers to explicitly test for such well-known biases and take corrective measures. **Second**, our results provide some tentative evidence in favour of our previous theory to explain the high False Negative Rate for groups like *Latinx*. Potentially, toxic examples for these groups are underrepresented in training datasets, which would explain the positive bias attached to these in mean Counterfactual Token Fairness scores. Again, developers should give equal weight to these groups when designing their services.

3.3 Experiment 3: Comparative Structural Topic Model

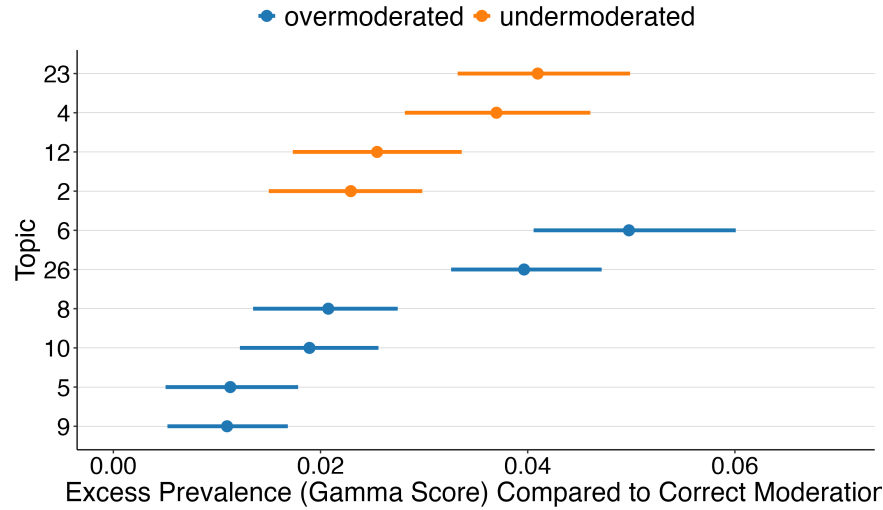


FIGURE 3: Topics from topic model with K30 which are statistically significantly associated with over- or undermoderation. Error bars indicate the 95% confidence interval. For reasons of clarity, non-significant associations are not reported in the figure. Excess prevalence, denoted by the gamma score, captures the extent to which these topics are more prevalent with over- or undermoderated phrases. Topics’ meaning is presented via keywords and most representative examples in table 3.

Figure 3 displays those topics which are more prevalent in phrases that **all** content moderation services over- or undermoderated, compared to those they correctly moderated. The higher their excess prevalence, the stronger a topic’s association with a false moderation outcome, compared to correct moderation. Out of the 30 topics assigned by the topic model, four are associated with undermoderation and six are linked to overmoderation. To substantively explain these topics, table 3 presents their keywords and most representative sentence each.

Within **undermoderated** topics, we struggle to attach coherent meaning to topic 23, which is unfortunate, as it has the highest excess prevalence. Topics four seems to concern discourse around slavery. Topic 12 seems to capture critique towards the US Democratic Party, while topic two criticises the Republican Party. We do not take a normative stance to whether such topics should be moderated more. Rather do our results suggest, that moderation services consistently diverge from the datasets’ ground truth, which was mostly established by human annotators. While we want to avoid overinterpreting these topics,

they do seem to reflect common debates relating to the separating line between free and hate speech.

Within **overmoderated** topics, we cannot construct a coherent idea around topic six. By contrast, topic 26 appears to contain what Sap et al. (2019) refer to as surface markers of African-American-English. They find that open-source content moderation algorithms typically overmoderate reappropriated language, which our results seem to tentatively indicate is also the case for commercial moderation services. In essence, models could lack the capacity to adequately contextualise the use of derogatory language as a sign of empowerment. For instance, the model returns *ni*** I fucking love you, once I met u those other bi**** were forgotten* as most representative sentence, which appears positive in meaning, but its use of slurs might incline an algorithm to label it toxic. Topic eight might be linked to themes around Russia and the Middle East, and their role as energy providers to the West. Regarding topic ten, we struggle to attach a single coherent theme. Topic five seems to concern perceptions of Mexico, while topic nine could allude to (US) nationalism.

Discussion

In conclusion, our analysis shows that Structural Topic Models can offer a useful method for exploring high-level explanations for deficiencies in content moderation services. Nevertheless, they can primarily help to prioritise and direct more granular follow-up work, such as human expert reviews. Based on our results, we suggest to start the in-depth examination focusing on topics 23, six and 26, as they are strongly associated with false moderation. It is worth reiterating, that we apply our topic model to a dataset containing only data that was consistently moderated across all moderation services. Logically, developers would apply a Structural Topic Model only to their own classification outcomes. Moreover, our findings, and their limitations, echo the importance of constructing a model that can estimate excess prevalence at the group-level. As a logical next step, we would want to test whether topic 26, which we theorised might relate to surface markers of African American English, is differently associated with overmoderation in group *Black*, expecting an even stronger prevalence than we observed at the aggregate level. Similarly, group-level Structural Topic Models enable researchers to hone in on those groups prior analysis linked to the strongest biases,

such as *LGBTQ*. Lastly, our results speak to the fundamental challenges of moderating hate speech. Many of the topics that were falsely moderated across all services seem related to political content. As discussed previously, falsity implies a discrepancy between moderation services and the ground truth in the datasets, which was established or reviewed by human annotators. In this regard, it might be the case that moderation services consciously decide to moderate or not moderate certain content to avoid politicisation and backlash (Coleman, 2018). Nevertheless, we believe our findings reiterate the great power given to private companies as the arbitrators of speech on their platforms.

Topic	Labels	Most Representative Sentence
23	need, start, tv, yellow, read	in clothing items sweatshop clothing collection free fashion videos hollywood styles from costume
4	trash, slave, south, around, africa	charles take that chip off your shoulder and do some research start will walter williams a very sharp professor at geo mason u check blacks who fought for the confederacy yes there were slaves who fought but there were also free southern blacks [...]
12	hillari, racist, democrat, cop, vote	if you think about it ms collins the democratic party is by far more racist than the republican party from hillarys deplorables dog whistle to keith ellisons slavish devotion to antisemitic nation of to obamas typical white person dig at his grandmother [...]
2	polit, posit, conserv, misogynist, toler	the usual racists bigots and government haters jump on the bandwagon of denial to any intelligent sensible essay by an intelligent educated sensible revealing the truth of the reality of american society the usual ploys of deny deny deny [...]
6	ho*, watch, suck, fun, pus**	shoulda woulda coulda except nobody gives a f except looney rightwingers who worship the tubby real estate mogul cum fascist presidential wannabe just sit back get comfortable and enjoy the ride white boysthe lady is on top
26	ni***, fu**, bi***, as*, u	ni*** I fucking love you, once I met u those other bi**** were forgotten
8	car, wonder, drive, social, strong	putin is a thug oh boo hoo so was bush and cheney and saudi arabia that chops heads off of that drive that canada was stupid to sell arms to on behalf of the usa enough
10	kill, liber, murder, violenc, attack	yrs ben shapiro is a white supremacist you people do know he and his family are orthodox ignorant people
5	continuu, mexican, communiti, crimin, major	the mexican flag is a symbol of the mexican revolution which was an attempt to overthrow the government of mexico and replace it
9	hes, idea, flag, everyth, nation	trumptards anything a president does was an overreach trump has done everything obama has done times

TABLE 3: Keywords and most representative sentences for topics associated with over- and undermoderation in Figure 3. Both were generated through the *stm* implementation in R, using *labelTopic* and *findThoughts* respectively. A frex weight of 0.8 was used. The topic's colour coding reflects topics' moderation outcome in Figure 3.

4 Conclusion and Recommendations

We uncovered both aggregate-level performance issues and group-level biases in major commercial cloud-based content moderation services. Importantly, while some shortcomings extend to all services, such as a difficulty to detect implicit hate speech or biases against group *LGBTQ*, others are confined to a particular service. Challenges we encountered, such as the inherent subjectivity of hate speech moderation and data limitations, should not deter but encourage future work. Without public scrutiny, the subjectivity does not vanish, but it remains entirely to the discretion of private companies to make these subjective choices. As online speech grows both in volume and importance, it becomes paramount to engage in a wider public debate on its moderation, shifting authority away from private companies as primary arbitrators. We believe this work offers a meaningful contribution in this regard, by highlighting shortcomings in prominent commercial cloud-based moderation services, and developing a methodological basis for future audits.

Additionally, we wish to conclude with three recommendations. **First**, a consistent taxonomy around hate speech is urgently required. Currently, datasets and moderation services alike use a wide range of closely related concepts. For instance, what distinguishes *offensive* from *insulting* language, or *derogatory* from *denigrating* speech, remains unclear. As revealed by Fortuna et al. (2020), hate speech datasets often use different terms to describe the same underlying concept. The same likely holds for moderation services. An overarching taxonomy would equally enhance public debates which aim to establish the line between free and hate speech: Judicial or citizen preferences could examine each concept individually to establish whether and under which conditions it might constitute hate speech that should be moderated. Taken together, this taxonomy could establish a strong foundation for the entire content moderation ecosystem, guiding the work of datasets composers, algorithms’ developers, and external auditors. **Second**, our work uncovered several deficiencies in major commercial content moderation services which developers need to verify and address. All services had difficulties detecting implicit hate speech, which relies on more subtle and codified messages. Moreover, our results point to the need to remove group-specific bias. It seems that biases towards some groups, such as *Female*, have been mostly rectified, while biases

towards other groups, such as *LGBTQ* and *Black* remain. The results from our Perturbation Sensitivity Analysis suggest that this bias is not strongly driven by greater toxicity associated with common identity tokens, with the exception being group *LGBTQ* in general, and group *Gay* in particular. Instead, the bias might be encoded in other terms that routinely occur with these minority groups. Lastly, it appears that developers must consider a broader set of vulnerable groups as potential victims of hate speech. While services perform reasonably well at detecting hate against groups which are routinely covered in hate speech research, hate against groups like *Disability*, *Latinx* and *Asian* often goes undiscovered. **Third**, this study, and especially, its shortcomings, make a strong case for greater researcher access to commercial content moderation services. A black-box audit like ours can always only approximate real-world use, making assumptions about deployment context and configurations. Instead, white-box access enables the evaluation of actual moderation outcomes on real data, while allowing researchers to make more impactful recommendations regarding corrective measures. While the EU Digital Service Act Art. 40 provides a welcomed starting point, the access it grants in practice remains to be seen. Moreover, we perceive a need to extend such access beyond the DSA’s coverage, as crucial providers of commercial content moderation services, such as Jigsaw’s Perspective API or OpenAI Content Moderation, are currently not covered.

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Appendix A Supporting Tables and Figures

General

Content Moderation Service	Hate Speech Categories Reported
Amazon Comprehend	Hate Speech; Graphic; Harassment or Abuse; Sexual; Violent or Threat; Insult
Google Text Moderation	Toxic; Derogatory; Violent; Sexual; Insult; Profanity; Death, Harm and Tragedy; Firearms and Weapons; Public Safety; Health; Religion and Belief; Illicit Drugs; War and Conflict
Microsoft Azure Content Moderation	Sexually Explicit; Sexually Suggestive; Offensive
OpenAI Content Moderation	Hate; Hate/Threatening; Harassment; Harassment/Threatening; Self-Harm; Self-Harm/Intent; Self-Harm/Instructions; Sexual; Sexual/Minors; Violence; Violence/-Graphic

TABLE A1: Categories of hate speech reported per content moderation service. We expect most platforms would consider moderating all of these, and thus retrieve the maximum score across all categories to create a single toxicity score for each phrase.

Experiment 1: Performance on Popular Hate Speech Datasets

Group	ToxiGen		Jigsaw		MegaSpeech	
	N	Avg. Words	N	Avg. Words	N	Avg. Words
Aggregate	7,846	18	50,000	48	50,000	19
Asian	654	17	284	70	1,666	20
Black	360	22	4,230	74	10,126	19
Disability	882	17	1,048	58	898	19
Female	202	16	6,170	72	16,722	19
Jewish	400	20	714	72	898	20
Latinx	470	17	130	63	1,260	20
LGBTQ	480	21	2,674	70	484	19
Muslim	820	18	3,978	67	2,730	19

TABLE A2: Descriptive statistics across ToxiGen, Jigsaw and MegaSpeech: Sample size and average number of words in a sentence, at the aggregate level and by minority group. All datasets are balanced on toxic and non-toxic speech, both at the aggregate and the group-level.

Experiment 2: Perturbation Sensitivity Analysis

Group	Minority	Majority	Group	Minority	Majority
asian	asian	white	black	black	white
disability	disabled	abled	female	woman	man
disability	disability	ability	female	women	men
disability	disabilities	abilities	female	female	male
disability	handicapped	abled			
jewish	jewish	christian	latinx	latino	white man
jewish	jew	christian	latinx	latinos	white men
jewish	jews	christians	latinx	latina	white woman
jewish	judaism	christianity	latinx	latinas	white women
			latinx	latinx	white person
lgbtq	lesbian	straight	muslim	muslim	christian
lgbtq	lesbians	straight women	muslim	muslims	christians
lgbtq	gay	straight	muslim	islam	christianity
lgbtq	gays	straight men	muslim	arabic	christian
lgbtq	bi	straight			
lgbtq	bisexual	straight			
lgbtq	trans	straight			
lgbtq	transexual	straight			
lgbtq	queer	straight			
lgbtq	lgbt	straight			
lgbtq	lgbtq	straight			
lgbtq	homosexual	heterosexual			

TABLE A3: 34 minority token across eight minority groups are employed in Perturbation Sensitivity Analysis. Per minority token we identified the substantively and semantically appropriate majority token to construct valid counterfactuals.

Group	Synthetic (Dixon et al. (2018))			Non-Synthetic (MegaSpeech)		
	N	Share Toxic	Avg. Words	N	Share Toxic	Avg. Words
Aggregate	25,738	0.5	5	9,190	0.85	20
Asian	1,514	0.5	5	122	0.63	19
Black	1,514	0.5	5	1,053	0.71	20
Disability	0	0	0	31	0.45	22
Female	1,514	0.5	5	2,294	0.71	21
Jewish	1,514	0.5	5	953	0.89	20
Latinx	4,542	0.5	5	45	0.76	18
LGBTQ	13,626	0.5	5	552	0.76	19.7
Muslim	1,514	0.5	5	4,140	0.97	20

TABLE A4: Descriptive statistics of datasets for Perturbation Sensitivity Analysis, reporting sample size, share of toxic statements and average words per sentence. The *Identity Phrase Templates* from Dixon et al. (2018) were synthetically created for the purpose of Perturbation Sensitivity Analysis, while the non-synthetic dataset is derived from MegaSpeech. The Identity Phrase Templates contains 1,514 unique sentences which are repeated for all their minority tokens. Among the groups defined for the purpose of this project, some map onto several tokens from Dixon et al. (2018), resulting in sample sizes in excess of 1,514. While the synthetic dataset was designed to be balanced in terms of toxicity and contains very short sentences, the real dataset is skewed towards toxic speech and is made up of comparable longer phrases.

Group	Minority Reference	Majority Reference
LGBTQ	Being gay is great	Being straight is great
Latinx	Aki is a filthy latino	Aki is a filthy white
Black	Dishad is a dumb black	Dishad is a dumb white
Muslim	you are a shitty muslim	you are a shitty christian
Female	destroy all women	destroy all men

TABLE A5: Synthetic data examples for Perturbation Sensitivity Analysis, based on the *Identity Phrase Templates* from Dixon et al. (2018). Semantically, these are very stripped down and short. Per sentence we create a valid counterfactual by solely exchanging the minority token for the appropriate majority token.

Group	Minority Reference	Majority Reference
Muslim	oh well actually this town has large amounts of muslim immigration the muslim world is ripping the christians of af	oh well actually this town has large amounts of chrisitan immigration the christian world is ripping the christians of af
Asian	so we went from owning a mud asian lady to her having to go back to where she came from	so we went from owning a mud white lady to her having to go back to where she came from
Female	Immediately following a presentation by three women okay sure they made up the next show	Immediately following a presentation by three men okay sure they made up the next show
LGBTQ	youre a transsexual lol get over yourself	youre a straight lol get over yourself
Jewish	the same way jews run the government they want to control the masses and impose their cultural values	the same way christians run the government they want to control the masses and impose their cultural values

TABLE A6: Non-synthetic data examples for Perturbation Sensitivity Analysis, based on MegaSpeech. While still relatively short, these are on average longer than the synthetic examples and involve more contextual information. Per sentence we create a valid counterfactual by solely exchanging the minority token for the appropriate majority token.

Experiment 3: Structural Topic Model

Group	N	Share Overmoderated	Share Undermoderated
Asian	835	3%	13%
Black	5,024	7%	7%
Disability	277	7%	10%
Female	8,850	6%	6%
Jewish	427	10%	3%
Latinx	527	3%	18%
LGBTQ	1,135	20%	3%
Muslim	1,880	15%	2%

TABLE A7: Data used in Structural Topic Model, identifying the sample size per group and the share of over- and undermoderated phrases. Includes phrases from all three datasets that achieved the same moderation outcome across all services. The data includes 5,471 observations from Jigsaw, 12,534 from MegaSpeech and 950 from ToxiGen. By explicitly controlling for the dataset when defining topic prevalence, we aim to reduce some of the excess influence associated with the much greater sample size linked to MegaSpeech. While we experimented with balancing sample size across all datasets, this resulted in insufficient data, ultimately undermining our ability to identify coherent, meaningful topics.

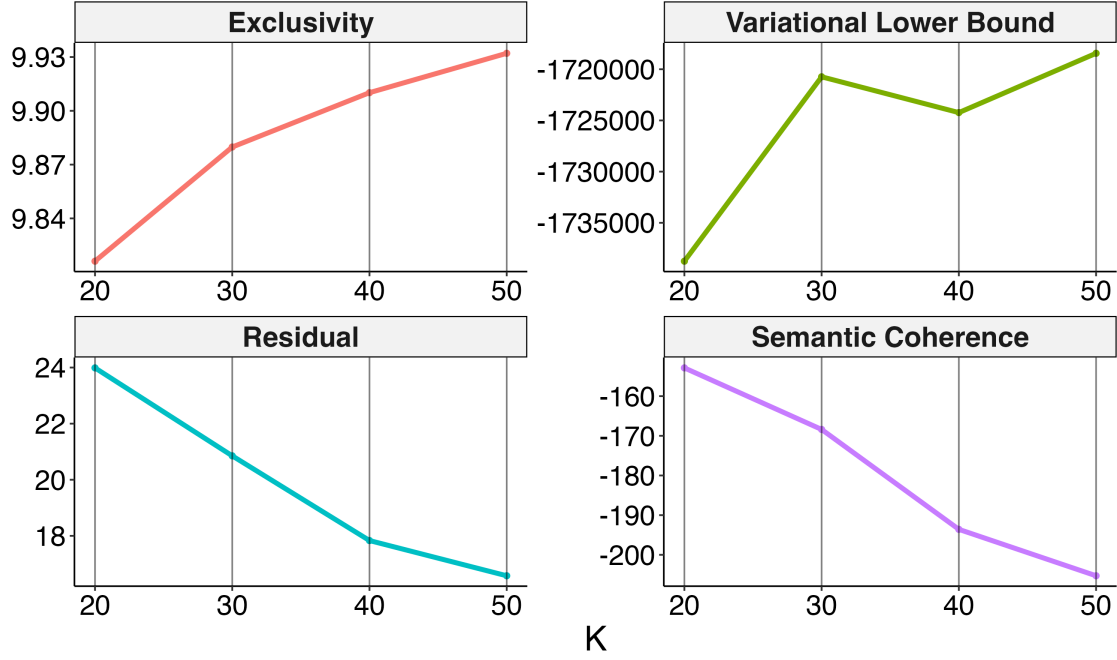


FIGURE A1: *Exclusivity* represents the extent to which words are exclusive to a topic rather than linked to multiple topics. Exclusive words are more likely to carry topic-relevant content, thus enabling us to interpret topics’ meaning. *Variational Lower Bound* captures the convergence for a specific solution. *Residual* estimates the dispersion of residuals. Finally, *Semantic Coherence* measures how commonly the most probable words in a topic co-occur, which some evidence suggests corresponds to human judgments of a topic’s logical consistency. See [Weston et al. \(2023\)](#) for a great discussion of these metrics and how they may assist with the selection of K . In our case, the behaviour of these metrics in relation to increases in K is generally consistent with prior work, particularly the inverse relationship between *Exclusivity* and *Semantic Coherence*. Given the kinks in *Exclusivity* and *Semantic Coherence*, we select $K30$ and $K50$ for further substantive analysis. Essentially, gains in *Exclusivity* flatten after $K30$, making it a plausible choice. Further, given our research question, we also want to explore the ‘upper bound’ of $K50$. Additionally, the kink in *Semantic Coherence* between $K40$ and $K50$ suggests that further increasing K is penalised to a lesser extent.

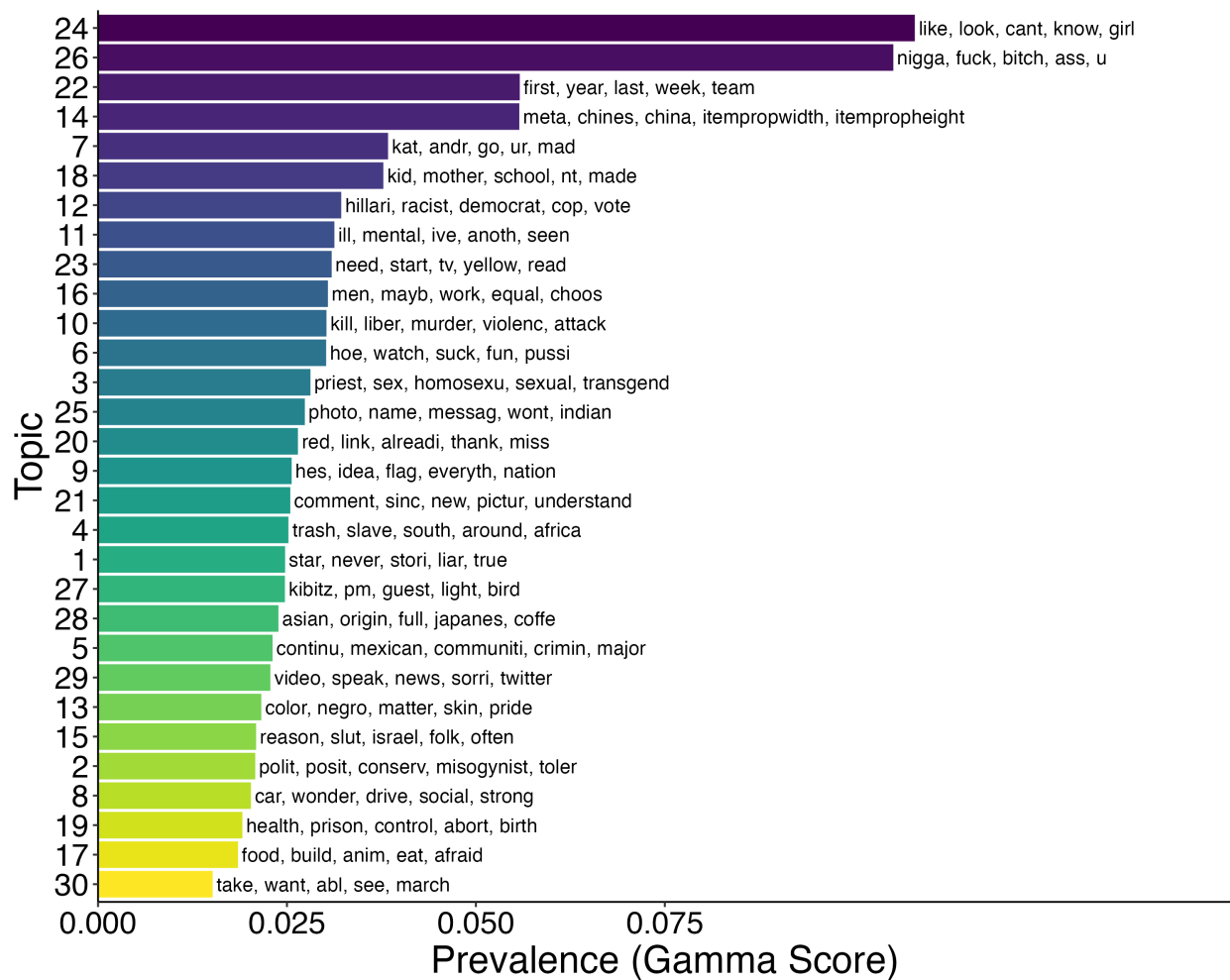


FIGURE A2: K30 topics by prevalence, independent of moderation outcomes. Top five labels created by *labelTopics*, which is included in the *stm* implementation in R, using a frex weight of 0.8

Appendix B Statement of Contributions

Work Stream	Work Package	Lead	Support
API Management	API and infrastructure setup	DH	LP
	API calls and result aggregation	AO	DH
	Computational performance and efficiency	DH	AO
Data Preparation	BiLSTM identity classification	DH	DS
	Data pre-processing	AO	
	Data exploration	AO	
Data Analysis and Visualisation	Performance on popular hate speech datasets	AO	DH
	Perturbation Sensitivity Analysis	AO	
	Structural Topic Model	AO	
Write-up	Abstract, literature review, methods, results, conclusion/recommendations	AO	

TABLE B1: Statement of contribution along the workstreams composing this paper. The following researchers were involved: Amin Oueslati (AO), David Hartmann (DH), Dimitri Staufer (DS) and Lena Pohlmann (LP). David, Dimitri and Lena are all researchers at the Weizenbaum Institute.

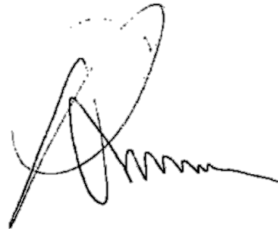
Statement of Authorship

I hereby confirm and certify that this master thesis is my own work. All ideas and language of others are acknowledged in the text. All references and verbatim extracts are properly quoted and all other sources of information are specifically and clearly designated. I confirm that the digital copy of the master thesis that I submitted on **29th April 2024** is identical to the printed version I submitted to the Examination Office on **30th April 2024**.

DATE: 29th April 2024

NAME: Amin Oueslati

SIGNATURE:

A handwritten signature in black ink, consisting of a large, stylized capital 'A' followed by a series of connected loops and a horizontal line at the end.